

## Introduction to Threat hunting

### 1- Definition

Threat hunting is a proactive way to find security threats. Instead of waiting for an alert, security experts actively search through network and computer logs to catch hidden hackers that traditional tools might miss.

There are three main ways to do threat hunting:

- **Hypothesis-Driven:** Experts assume a hacker is already inside the system and search to see if they are hiding there.
- **Intelligence-Driven:** Experts use information about how hackers usually attack to find clues left behind on the network.
- **Analytics-Driven:** Experts use smart tools and machine learning to spot unusual behavior, like someone logging in at 3:00 AM when the office is closed.

### 2- Effective Log Collection Strategies for Threat Hunting with Wazuh

#### A- Log data collection for threat hunting.

Standard logs are not enough for threat hunting. We need deep visibility into logs, like which process has been run, who established a network connection, etc. This means we collect high-quality logs for threat hunting.

#### B- The difference between standard logging and threat hunting logging.

| Feature                       | Source                                   | Focus                                                         | Volume                    | Goal                                        |
|-------------------------------|------------------------------------------|---------------------------------------------------------------|---------------------------|---------------------------------------------|
| <b>Standard Logging</b>       | Windows Security Events, Syslog          | Login failure, System errors                                  | Low to Medium             | Compliance & Basic Security                 |
| <b>Threat Hunting Logging</b> | Sysmon, Auditd, PowerShell Script Blocks | Process creation, Command line arguments, Network connections | High (Requires filtering) | Detecting sophisticated attacks & Anomalies |

Logs are collected based on four features: **source, focus, volume, and goal.**

**In terms of source:** standard logs collect Windows security logs or Syslogs and threat hunting point of view, we collect logs or sources like Sysmon, Auditd, and PowerShell scripts.

**In terms of focus:** standard logging includes logs like login failures, system errors, or basic logs and threat hunting point of view includes process creation, command line arguments, network connections, etc.

**In terms of volume:** Volume is low to medium for standard logs, and in threat hunting, we have a huge volume of HD hard drive data, from which we have to filter logs according to our requirements.

**In terms of goal:** The basic purpose of standard logging is to ensure compliance or basic security, but in threat hunting, we have to detect persistent, deep, sophisticated attacks or anomalies or APTs.

### 3- Preparation the environment for threat hunting.

#### A- Let's start with Windows first.

The most powerful tool for Windows threat hunting is Sysmon, because default Windows logs normally do not show process creation details like parent process and command line in detail. In the previous course, we configured Sysmon.

Now we will collect our Sysmon operational logs.

So go to Agent Management → Groups → open the configuration file for specific group (here we chose our windows group)

Add this configuration block for your Sysmon logs.

```
<!-- Enable Sysmon Logs -->
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Now we will enable our PowerShell logging. It's optional but recommended

Add its configuration block

```
<!-- Enable PowerShell Logs -->
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

After that, click on Save.

#### B- Do the same to set up threat hunting on our Linux.

For threat hunting on Linux, it is very important to enable Auditd and bash command logging.

Run the command: `sudo apt-get install auditd audispd-plugins -y` and hit Enter, to install the service.

To ensure that whenever our Linux machine restarts, the audit logs restart automatically, run this command: `sudo systemctl enable auditd`

Go into our Linux group and enable the audit logs by adding this configuration block:

```
<!-- Enable Auditd Logs -->

<localfile>
<log_format>audit</log_format>
<location>/var/log/audit/audit.log</location>
</localfile>

<localfile>
<log_format>syslog</log_format>
<location>/var/log/auth.log</location>
</localfile>
```

After that, click on Save and restart our Wazuh server.

Navigate back to the Wazuh dashboard to review incoming alerts, click on **Low severity logs**, select your Windows desktop agent, and filter by the provider name `(data.win.system.provider.Name)` to confirm that Sysmon logs are actively collecting.

|   |                      |   |   |                 |   |   |                          |
|---|----------------------|---|---|-----------------|---|---|--------------------------|
| > | Sep 13, 2026 @ 15:19 | + | - | DESKTOP-J20QDIK | + | - | Microsoft-Windows-Sysmon |
| > | Sep 13, 2026 @ 15:19 | + | - | DESKTOP-J20QDIK | + | - | Microsoft-Windows-Sysmon |
| > | Sep 13, 2026 @ 15:17 | + | - | DESKTOP-J20QDIK | + | - | Microsoft-Windows-Sysmon |

Next, switch to your Linux endpoint agent and filter by rule groups (`rule.groups`) to verify that Linux audit logs are streaming correctly.

|   |                      |   |   |                                     |   |   |           |   |   |
|---|----------------------|---|---|-------------------------------------|---|---|-----------|---|---|
| > | Sep 13, 2026 @ 15:26 | + | - | pam, syslog                         | + | - | ubuntu-01 | + | - |
| > | Sep 13, 2026 @ 15:26 | + | - | pam, syslog                         | + | - | ubuntu-01 | + | - |
| > | Sep 13, 2026 @ 15:26 | + | - | pam, syslog, authentication_success | + | - | ubuntu-01 | + | - |
| > | Sep 13, 2026 @ 15:26 | + | - | syslog, sudo                        | + | - | ubuntu-01 | + | - |

Successfully viewing and filtering logs from both multi-platform agents confirms that your SIEM telemetry pipeline is fully operational for threat hunting.



**WAZUH**  
Open Source Security  
Monitoring Platform

# Introduction to Threat Hunting with Wazuh (Windows & Linux)

Collect Logs Analyze Hunt Threats Detect Anomalies Respond



"Knowledge  
Builds a Safer  
Tomorrow"  
— Tims TCHOUAMOU  
Cybersecurity Enthusiast

## 1 What is Threat Hunting?



Threat hunting is a proactive way to find security threats. Instead of waiting for an alert, security experts actively search through network and computer logs to catch hidden hackers that traditional tools might miss.

Three main approaches:



**Hypothesis-Driven**  
Experts assume a hacker is already inside the system and search to see if they are hiding there.



**Intelligence-Driven**  
Experts use information about how hackers usually attack to find clues left behind on the network.



**Analytics-Driven**  
Experts use smart tools and machine learning to spot unusual behavior, like someone logging in at 3:00 AM when the office is closed.

"Be proactive. Find what others miss."

## 2 Why Good Log Collection Matters?



Standard logs are not enough for threat hunting. We need deep visibility into logs, like which process has been run, who established a network connection, etc. This means we collect high-quality logs for threat hunting.

| Feature | Standard Logging                | Threat Hunting Logging                                        |
|---------|---------------------------------|---------------------------------------------------------------|
| Source  | Windows Security Events, Syslog | Sysmon, Auditd, PowerShell Script Blocks                      |
| Focus   | Login failure, System errors    | Process creation, Command line arguments, Network connections |
| Volume  | Low to Medium                   | High (Requires filtering)                                     |
| Goal    | Compliance & Basic Security     | Detecting sophisticated attacks & Anomalies                   |



Logs are collected based on four features: source, focus, volume, and goal. Threat hunting requires more detailed logs, a higher volume of data, and a focus on detecting persistent and advanced threats (APTs).

## 3 Centralized Group Configuration



We use centralized group configuration (Windows group) so that when we push changes from the Wazuh manager, it will apply those changes to every Windows agent.

- 1 Log in to the Wazuh dashboard.
- 2 Go to **Agent Management** → **Groups**.
- 3 Select your Windows group (or create one).
- 4 Edit the agent configuration file (**agent.conf**).



Any configuration added to a group will be applied to all agents in that group.

## 4 Download Required Files (Windows)

- 1 Download Sysmon from Microsoft Sysinternals:  
<https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

Microsoft | Learn | Sysinternals | Downloads



- 2 Download the Olaf Hems XML configuration file:  
<https://github.com/tim87tchoua/sysmon-modular/blob/master/sysmonconfig.xml>



## 5 Prepare and Install Sysmon

- 1 Extract the Sysmon zip file.
- 2 Copy the downloaded sysmonconfig.xml into the Sysmon folder.
- 3 Move the Sysmon folder to the C: drive (C:\Sysmon).
- 4 Open Windows PowerShell as Administrator.
- 5 Navigate to the Sysmon folder and install Sysmon:

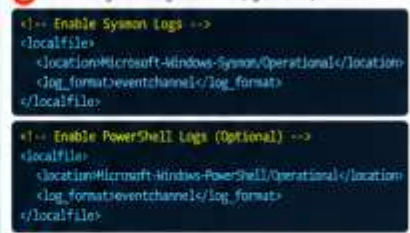


Sysmon will be installed on the endpoint.



## 6 Configure Wazuh Agent (Windows)

- 1 In the Wazuh dashboard, go to **Agent Management** → **Groups** → **Windows group**.
- 2 Edit the agent configuration file (**agent.conf**) and add:



- 3 Save the configuration and restart the Wazuh manager.

PowerShell logging is optional but recommended for better visibility.

## 7 Prepare Linux Environment (Auditd)

- 1 Install auditd and plugins:  
`tim@bunbu:~$ sudo apt-get install auditd auditd-plugins -y`

- 2 Enable the service at boot:  
`tim@bunbu:~$ sudo systemctl enable auditd`

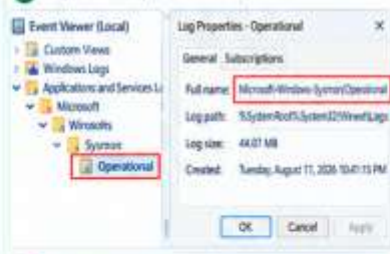
- 3 In the Wazuh Linux group, add this configuration block:



- 4 Save the configuration and restart the Wazuh manager.

## 8 Find the Event Channel Name

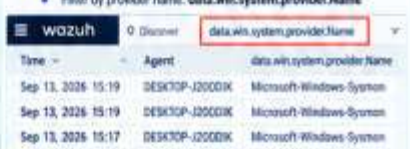
- 1 On the Windows agent, open **Event Viewer**.
- 2 Navigate to: **Application and Services Logs** → **Microsoft** → **Windows** → **Sysmon** → **Operational**.
- 3 Right-click on **Operational** → **Properties**.



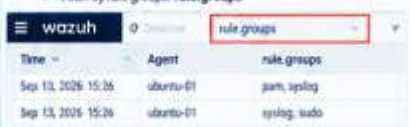
Use the full name: **Microsoft-Windows-Sysmon/Operational**

## 9 Verify Logs in Wazuh Dashboard

- 1 Check Windows Sysmon logs:  
• Go to Wazuh dashboard → **Security events**.  
• Select your Windows agent.  
• Filter by provider name: **data.win.system.providerName**



- 2 Check Linux audit logs:  
• Select your Linux agent.  
• Filter by rule groups: **rule.groups**



## 10 Result: SIEM Ready for Threat Hunting



Successfully viewing and filtering logs from both multi-platform agents confirms that your SIEM telemetry pipeline is fully operational for threat hunting.

## Key Takeaways

- Collect high-quality logs (Sysmon, Auditd, PowerShell).
- Use centralized group configuration.
- Monitor process creation, command lines, and network connections.
- Filter and analyze logs to detect anomalies.
- Be proactive, hunt threats before they cause damage.

## Author & Motivation



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Cybersecurity Enthusiast | SOC Content Creator  
Learn • Practice • Secure • Share

"Knowledge Builds a Safer Tomorrow"



**WAZUH**  
Open Source Security  
Monitoring Platform



Stranger Defenders.  
A Safer Tomorrow.